Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/18/2024
PI Name:	Dmochowski, Jacek Ph.D.		
Project Title:	Boosting Brain Metabolism in Spaceflight with Transcranial Photobiomodulation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031-9129	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	08/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/30/2021 per E. Urquieta/TRISH (Ed., 6/9/21) NOTE: End date changed to 06/30/2021 per TRISH (Ed., 7/6/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liu, Hanli Ph.D. (The University of Texas System)		
Grant/Contract No.:	NNX16AO69A-T0409		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this proposal is to develop a new technique for enhancing brain activity by applying safe levels of laser light to the forehead. It is known that near-infrared light is absorbed by an enzyme in the mitochondria and subsequently leads to greater energy availability in the stimulated cells. When stimulating the brain, this additional energy can be used by brain cells to carry out the necessary functions required to maintain high cognitive performance (for example, sustained attention, accurate memory recall). We will test the proposed technology by scanning participants' brain activity while they receive laser stimulation to the forehead and at the same time perform a challenging perceptual task. We will also perform experiments where Light Emitting Diodes (LEDs) are used to stimulate participants' brains while they are trained on a working memory task and simultaneously have their brain activity measured. Our expectation is that we will observe increased energy metabolism in the frontal part of the brain, as well as an improvement in the performance of both perceptual and memory tasks. We expect that this research will lead to a new, non-pharmacological technique for boosting the performance of the brain by delivering safe levels of near-infrared light to the brain.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology being developed has potential utility in spaceflight, investigative treatments for depression and other vascular related disorders, as well as cognitive enhancement. We expect that the publications stemming from this project will become highly cited and spur the development of transcranial photobiomodulation (tPBM) in basic and clinical neuroscience.
Task Progress:	(1) Original project aims/objectives. The objective of this proposal is to develop a new technique for enhancing brain activity by applying safe levels of laser light to the forehead. It is known that near-infrared light is absorbed by an enzyme in the mitochondria and subsequently leads to greater energy availability in the stimulated cells. When stimulating the brain, this additional energy can be used by brain cells to carry out the necessary functions required to maintain high cognitive performance (for example, sustained attention, accurate memory recall). We will test the proposed technology by scanning participants' brain activity while they receive laser stimulation to the forehead and at the same time perform a challenging perceptual task. We will also perform experiments where Light Emitting Diodes (LEDs) are used to stimulate participants' brains while they are trained on a working memory task and simultaneously have their brain activity measured. Our expectation is that we will observe increased energy metabolism in the frontal part of the brain, as well as an improvement in the performance of both perceptual and memory tasks. We expect that this research will lead to a new, non-pharmacological technique for boosting the performance of the brain by delivering safe levels of near-infrared light to the brain. (2) Key findings We have demonstrated with MR Thermometry that transcranial photobiomodulation (tPBM) does not heat the brain tissue. To show this, we recruited n=20 healthy participants to receive both active and sham tPBM while recording their brain temperature with an MRI technique that is sensitive to temperature changes. We did not detect any time points of significant deviation in temperature between active and sham, and temperature increases were limited to 0.15 degrees Celcius, which is well within the range of physiological fluctuation. Therefore, we are confident that the technique is safe, and its mechanism does not rely on heating of the brain. We have also demonstrated a robust effect on functional connectivity in the human brain during tPBM. We scanned n=20 healthy participants with BOLD-fMRI, and then measured the functional connectivity among brain regions before, during, and after illumination. We found a 20-27% increase in connectivity during stimulation. This suggests that brain activity becomes synchronized during tBPM, pointing to its potential in improving cognitive performance. In collaboration with NYU Radiology, we have designed an Phosphorus MRS experiment with data collection beginning on December 18th. We have already collected pilot data without the tPBM to verify that we are able to resolve the important energy metabolites: PCr, three ATP groups. We are confident that we will be able to resolve a potential effect on ATP concentration with the current MRS sequence. UTA group: we have began collecting data to test the hypothesis in Aim 2 (11 subjects in both sham and active groups). Subjects have been performing the Psychomotor Vigilance Task and 2-back Tasks as implemented by the NASA Cognition software (courtesy of Mattias Basner). Participants receive tPBM with an LED. As this is a 12-week experiment, results are not yet available. (3) Impact of key findings on hypotheses, technology requirements, objectives and specific aims of the original proposal Our findings have been positive and lend further credence to our central hypothesis that tPBM is capable of enhancing cognitive function during spaceflight. We will continue with our research activities as planned in the proposal. We have strong evidence for the safety of our intervention, and also for the ability of near-infrared light to stimulate mitochondrial function in the human brain. We are very encouraged by the findings from the first year. (4) Proposed research plan for the coming year We will be scanning N=26 healthy participants between December 18th, 2019 and June 18th, 2020 at NYU with Phosphorus MRS. These experiments will produce data that will shed important light as to whether the mechanism of tPBM is indeed enhanced mitochondrial activity. We will continue the 12-week LED based study taking place at University of Texas Arlington. We will analyze data from both experiments, followed by dissemination of the findings in peer-reviewed journals.
Bibliography Type:	Description: (Last Updated: 03/13/2025)
Articles in Peer-reviewed Journals	Dmochowski GM, Shereen AD, Berisha D, Dmochowski JP. "Near-infrared light increases functional connectivity with a non-thermal mechanism." Cerebral Cortex Communications. 2020 Mar 19;1(1):tgaa004. https://doi.org/10.1093/texcom/tgaa004 ; PMID: 34296085; PMCID: PMC8152883 , Mar-2020